Installation Guide

Rescue Robot Markov Chain App

1 Requirements

Use Python 3.10, 3.11, or 3.12. The project was designed as a Streamlit application and should be run inside a Python virtual environment.

Required project files:

```
app.py
requirements.txt
utils/discrete.py
utils/continuous.py
utils/plotting.py
utils/policies.py
```

2 Option A – Install from a ZIP File

1. Download the project ZIP file.
2. Extract it to a folder of your choice.
3. Open a terminal inside the extracted project folder.

On Linux or macOS:

```
unzip Grupo1_app.zip
cd Grupo1_app
python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
streamlit run app.py
```

On Windows PowerShell:

```
Expand-Archive Grupo1_app.zip
cd Grupo1_app
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install -r requirements.txt
streamlit run app.py
```

After running the last command, Streamlit prints a local URL. Open it in a browser if it does not open automatically.

3 Option B – Install from GitHub

Clone the repository and install the dependencies:

```
git clone https://github.com/up202109000/markov.git
cd markov
python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
streamlit run app.py
```

On Windows PowerShell:

```
git clone https://github.com/up202109000/markov.git
cd markov
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install -r requirements.txt
streamlit run app.py
```

4 What the App Does

The application models a rescue robot moving in a grid with safe cells, hazards, and a victim cell. The mission is analyzed in two ways:

- A discrete-time Markov chain, where each robot movement is one step.
- A continuous-time Markov chain, where transitions happen after random waiting times.

The app computes and visualizes transition matrices, communication classes, n-step state distributions, absorbing-chain quantities, expected steps until absorption, success/failure probabilities, generator matrices, and continuous-time probability evolution.

5 Dependency List

The recommended `requirements.txt` is:

```
streamlit>=1.31,<2.0
numpy>=1.26,<3.0
pandas>=2.0,<3.0
scipy>=1.11,<2.0
plotly>=5.18,<7.0
pyvis>=0.3.2,<0.4.0
networkx>=3.2,<4.0
```

The current implementation does not require Gymnasium at runtime because the grid and transition model are generated directly by the project utilities. Gymnasium is only part of the conceptual FrozenLake inspiration.

6 Common Problems

6.1 `streamlit: command not found`

The virtual environment is probably not active, or Streamlit was not installed. Activate the environment and reinstall requirements:

```
source .venv/bin/activate
pip install -r requirements.txt
```

On Windows:

```
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
```

6.2 PyVis Transition Graph Does Not Appear

Install or reinstall PyVis:

```
pip install pyvis
```

Then restart Streamlit.

6.3 Browser Does Not Open Automatically

Copy the local URL printed by Streamlit, usually `http://localhost:8501`, and open it manually in the browser.

6.4 Dependency Conflict

Create a clean environment and reinstall:

```
rm -rf .venv
python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

On Windows, delete the `.venv` folder manually and recreate it.